

Nayem Sarkar

SOFTWARE ENGINEER | BACKEND ARCHITECT

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🌐 port-folio-blush-eta.vercel.app • 🏠 hackerrank.com/profile/nsarkar203032

PROFESSIONAL PROFILE

Abstract—Backend Engineer with a focus on designing high-throughput distributed systems and type-safe modular architectures. Expert in transforming complex legacy monoliths into maintainable services using Software Engineering principles. Proven track record in optimizing data integrity through rigorous schema engineering and architecting scalable server-side environments across diverse tech stacks.

TECHNICAL SKILLS

Backend Development

TypeScript, Node.js, NestJS, Express.js, Java, SpringBoot Python, C++

Systems & Data Architecture

Microservices, RabbitMQ, Redis, PostgreSQL, MySQL, MongoDB, Prisma, Knex, TypeORM

API, Security & Integrations

RESTful APIs, Swagger (OpenAPI), JWT/Passport.js, WebSockets, Stripe API

Cloud & DevOps

Docker, GitHub Actions, Git, Cloudinary, Render, CI/CD Pipelines, Bash Scripting

Testing & Quality Assurance

Black Box, White Box, Unit Testing, System Modeling

ENGINEERING EXPERIENCE

Betopia Group

Dhaka, Bangladesh

Software Engineer (Backend) | Internship

Dec 2025 – Apr 2026

- **Microservices Migration:** Refactored a legacy monolith into a **modular NestJS architecture**, reducing system coupling and enabling 3x faster feature iteration.
- **Recursive Logic Systems:** Architected a **tree-traversal progression engine** for a Piano Academy; utilized recursive MySQL queries to enforce 100% compliance.
- **Schema Engineering:** Optimized **multi-tenant relational schemas** using Prisma ORM, improving query response times by 25% for high-concurrency traffic.
- **Financial Integration:** Implemented **Stripe API** with an idempotent webhook architecture, ensuring 99.9% transaction reliability.

FEATURED PRODUCTS

VendoFoodvendofood.com

Multi-Vendor E-Commerce Platform

- Engineered a scalable multi-tenant architecture supporting concurrent storefronts.
- Implemented high-performance caching strategies using Redis for high-concurrency traffic.

SkтчLabsskтчlabs.com

Book Coloring E-Commerce

- Developed a custom digital asset delivery pipeline and automated print-on-demand triggers.

Witklip Farmwitklipfarm.com

Agricultural Service Platform

- Built an interactive service-scheduling platform using geospatial data integration.

TECHNICAL HONORS

Champion, System Analysis & DesignSpring 2024

Excellence in Software Architecture

 View

- Secured first place for the design of a scalable EdTech ecosystem architecture.

2nd Runner Up, Final Year Design ProjectSpring 2025

DeepQolYp: Hybrid Classical-Quantum Model

 View

- Developed a hybrid deep learning model utilizing quantum computing for medical imaging.

EDUCATION

United International University (UIU)Graduated 2025

Bachelor of Science in Computer Science & EngineeringDhaka, Bangladesh

- **Focus:** Data Structures, DBMS, Software Engineering, OS, System Analysis, OOP, Testing, SRS Documents, Business Analysis.

PROFESSIONAL REFERENCES

Jahidur Rahman

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